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ORIGINAL PAPER

Petrological and geochemical evidence for a hot crystallization path and a recharge filtering bypass at Antimilos, Milos volcanic field, Greece

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Abstract

Antimilos volcano in the South Aegean Volcanic Arc, Greece, comprises an andesite–dacite suite that follows a distinct evolutionary path than the main edifice of the Milos volcanic field, despite their proximity. Petrographic and geochemical analyses reveal that basaltic andesite to low-Si dacite lavas have similar phenocryst assemblages that indicate crystallization from hot, relatively dry magmas in an upper crustal storage region. Rare antecrystic high-Mg# clinopyroxene cores with low Y, low Dy, and high Sr contents record the cryptic involvement of amphibole, a phase nominally absent from the erupted products, in the deeper parts of the plumbing system. Low temperature antecrysts with textures recording various degrees of disequilibrium suggest a protracted history of interaction between the upper crustal reservoir and deeper mafic melts, forming mobile hybrid magmas that consequently erupt as highly mingled, crystal-rich lava domes. Antimilos magmas seem to have escaped recharge filtering in the upper crust and prolonged stalling, which is the process that is probably responsible for the paucity of mafic eruptions in the rest of the Milos volcanic system. Large extensional structures offshore of Antimilos promote rapid ascent of mafic melts, inhibiting prolonged stalling and interaction with the arc crust. This model highlights the dominant role of the regional stress field in generating petrologically distinct suites in the marginal parts of some volcanic fields.

Keywords Aegean arc · Milos volcanic field · Antimilos · Crystal cargo · Recharge filtering · Sub-volcanic plumbing system

Introduction

The volumetric dominance of andesite among the eruptive products in continental arcs (Gill 1981) has been explained in various ways. The most successful theories tend to involve mixing of mafic and felsic endmembers followed by preferential eruption of resultant hybrid magmas. Mixing, either of melts or of melt-crystal mixtures, is indicated by the often crystal-rich nature of intermediate rocks, the presence of variable crystal cargoes with abundant disequilibrium textures, and the scarcity of andesitic liquids in the melt inclusion record (Eichelberger 1978; Gill 1981; Eichelberger et al. 2006; Reubi and Blundy 2009). Felsic magmas are often too cold and crystal-rich to erupt, whereas mafic magmas are likely to stall during ascent due to their high densities (Plank and Langmuir 1988), and viscosity increases due to undercooling and crystallization (Pistone et al. 2013). Between these limits, there may be an optimum eruption

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potential for hybrid magmas (Sparks and Marshall 1986; Kent et al. 2010). The challenge of erupting either felsic or mafic endmembers and hence the key role of mafic recharge in bypassing thermal and rheologic barriers to eruption was termed "recharge filtering" by Kent et al. (2010). Magma mixing is promoted if the volatile content of the upper crustal reservoir is high enough to achieve water saturation and subsequent bubble exsolution, leading to a more compressible magma chamber where magmas can effectively mix for prolonged periods before eruption (Popa et al. 2021). This results in erupted hybrids with continuous compositions from basalt to rhyolite in continental arcs (Zhao et al. 2023). Effective hybridization during upper crustal storage may conceal any original bimodality, resulting in well-mixed bulk compositions, with scarce evidence for the nature of the mixing endmembers preserved in the crystal cargo of erupted hybrids (e.g., Streck et al. 2005; Eichelberger et al. 2006; Reubi and Blundy 2009; Kent et al. 2010). However, despite the important control on erupted compositions of many arc volcanoes attributed to recharge filtering, the mafic endmembers can and do occasionally erupt, often at some distance from the central volcanic edifice and commonly assisted by large extensional structures (e.g., Leeman et al. 1990; Cassidy et al. 2015).

Similar concepts have been applied to explain the range of erupted compositions at several of the volcanic centers that make up the active South Aegean arc (Fig. 1a). Mafic eruptions only occurred before the development of an upper crustal rhyolitic chamber in the large silicic system of Nisyros (Klaver et al. 2017; Popa et al. 2020a). At the smaller center of Methana, recharge filtering is thought to be most effective beneath the central edifice, effectively limiting the number of mafic eruptions onshore (Popa et al. 2020b), whereas nowadays mafic magmas ascend and erupt only along the northwestern flank, to form the submarine Pausanias volcanoes (Popa et al. 2020b; Woelki et al. 2022). In general, regional tectonic stresses and structures exert a strong influence on magma compositions at Methana (Pe-Piper and Piper 2013), resulting in eruptions of isotopically diverse products in response to changes in the regional stress field (Elburg et al. 2018). Similar radial structure is apparent in the diverse compositions of erupted magmas at Santorini. The marginal Akrotiri series and Kolumbo products exhibit ubiquitous signatures of amphibole fractionation, which are absent in the products of the central edifice (Mortazavi and Sparks 2004; Klaver et al. 2016; Francalanci and Zellmer 2019 and references therein).

This study focuses on the island of Antimilos (Fig. 1a, b), located on the NW flank of the ~ 3 Ma to present Milos Volcanic Field (MVF) (Fytikas et al. 1986; Francalanci et al. 2007; Zhou et al. 2021). By combining bulk rock geochemistry and a detailed analysis of texture and mineralogy of erupted units, we show that the Antimilos andesite–dacite suite records a hot evolutionary path. We argue that these magmas to some degree escaped cooling, and differentiation during upper crustal residence. The observed variations sample a wide range of different degrees of interaction between ascending recharge and resident felsic products. This contrasts with the main edifice of the MVF (Milos), where cooling and plutonic activity has been predominant over volcanism since the late Pliocene, likely due to localized compression (Zhou et al. 2022).

Regional geology

The island of Antimilos is located approximately 5 miles northwest of Milos Island (Fig. 1a) and is considered as the subaerial expression of the only composite volcano in the MVF (Fytikas et al. 1986; Vougioukalakis et al. 2019). The island covers an area of 8.5 km^2 with two main topographic highs, a main rounded peak (known as Sterna; Fig. 1c) of ~ 650 m elevation in the northern part of the island and a smaller peak (Agriokastro) of 413 m elevation in the south (Fig. 1d). Including the submarine topography, the entire dome-shaped edifice builds over 900 m of volcanic relief (Alexandri et al. 2001). It sits in the eastern end of a WNW-ESE neotectonic horst-graben structure that marks the westernmost extension of the South Aegean Arc (Nomikou et al. 2013). Early works from Sonder (1924) and Marinos (1961) briefly described the geology of the island and the petrology of the eruptive products. They report that most of the island consists of steep dacite lava domes, with subordinate andesite found only around the crater below the northern summit and in a few blocky lava flows around a smaller crater on the southern peak (Fig. 1d). Minor dacite flows also outcrop around the periphery of the northern crater (Sterna; Fig. 1c), where the products are mainly basaltic andesites, similar to the flows on the eastern slopes of the island. The only available age determination is a single K–Ar age of 0.32 ± 0.05 Ma on a whole-rock rhyolite (Fytikas et al. 1986). Based on field relations, Marinos (1961) indicated that the andesitic lavas of the northern and southern peaks of Antimilos are emplaced above the dacites and are presumably younger.

Materials and methods

We collected eleven samples from Antimilos during an excursion organized by a naturalist group from Milos, in September 2020. Samples are considered representative of the main eruptive centers defined in the geologic sketch of Marinos (1961), which was used as a sampling guide (Fig. 1b).

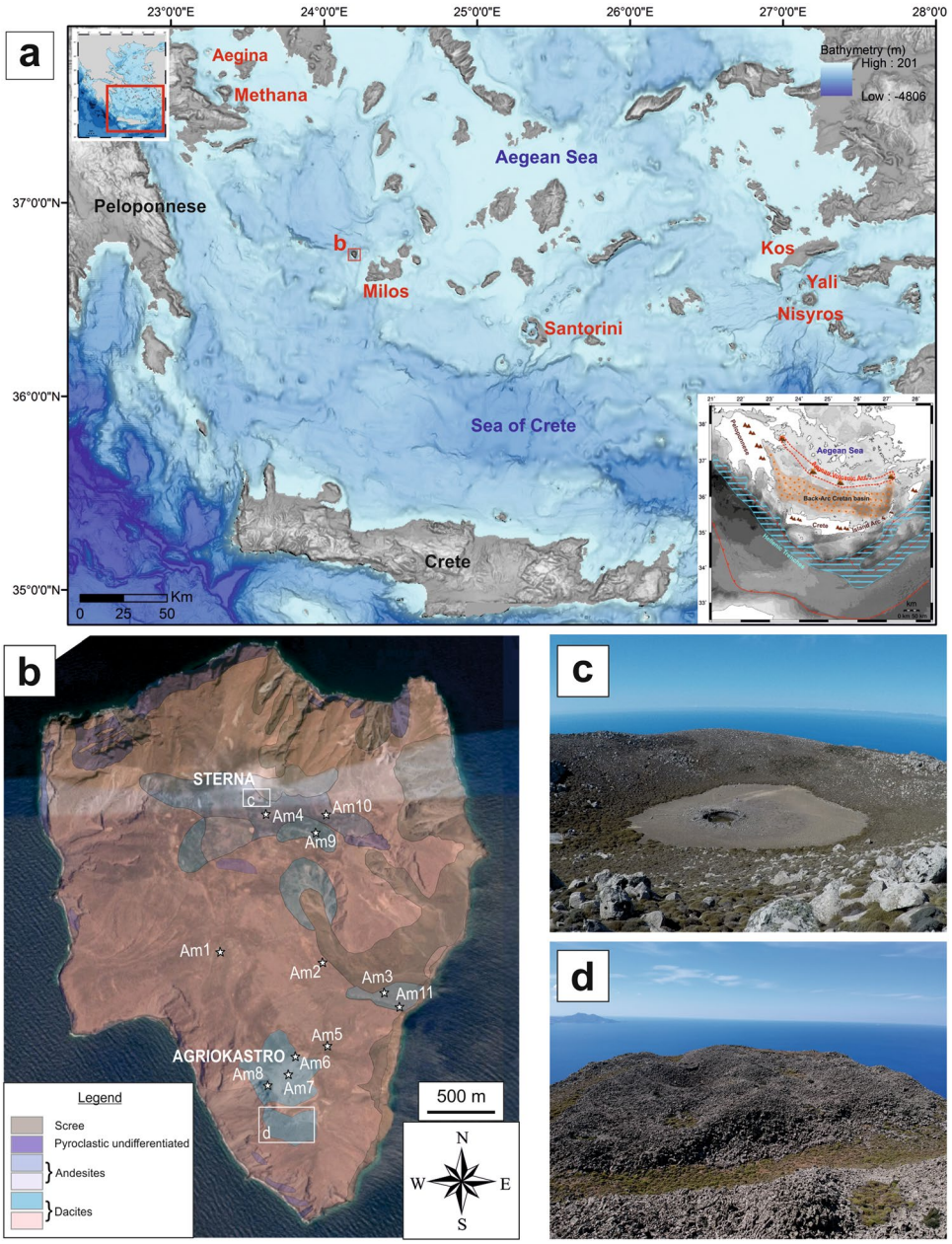


Fig. 1 **a** Regional bathymetric map of the Hellenic arc, back arc region with the major volcanic complexes of the South Aegean Volcanic Arc highlighted (after Nomikou et al. 2013), **b** Google Earth image of Antimilos island with the main outcropping units redrawn after the geological sketch of Marinos (1961), with the sampling points of the present study marked with the filled star symbols, **c** Overview of the crater lake beneath Sterna peak on the north part of Antimilos, **d** Blocky andesite flows with limited lateral extent on Agriokastro peak (South Antimilos)

Major element compositions of minerals were determined in eleven polished thin sections using a JEOL JXA8530F field emission microprobe at the Institut für Mineralogie, University of Münster (Germany). All analyses were performed with an accelerating voltage of 15 kV. For most minerals, a 20 nA focused beam current, 20 s counting time on peak position, and 10 s for each background were used. For glass and plagioclase, a slightly defocused (5 µm diameter) beam and 10 s counting time were used. Natural mineral standards used were albite (Na, Si, Al), wollastonite (Ca), olivine (Mg), almandine (Fe), spessartine (Mn), orthoclase (K), rutile (Ti), chromite (Cr), Ni-oxide (Ni) and Durango apatite (P). Compositions of pyroxenes, olivine, feldspar, biotite, and amphibole compositions are given in Supplementary data (Sheets 1–5).

Trace element analyses were carried out at University of Münster on a ThermoFisher Scientific Element 2 sector field ICP-MS coupled to a Photon Machines Analyte G2 Excimer laser system operating with ca. 5 J/cm^2 laser fluence and a repetition rate of 6–10 Hz. We used a large-volume ablation cell with fast signal response and short wash-out times (< 1 s) that holds up to six conventional thin sections and additional reference materials. Prior to sample analyses, the system was tuned with NIST SRM 612 glass for high sensitivity, stability, and low oxide-interference rates (^{232}Th^{16}O/^{232}Th < 0.2%). Spot sizes for the mineral analysis

were between 25 and 60 µm in diameter; in most cases 40 µm was selected as the best compromise between laser signal strength and spatial resolution. The signal ablation time was 40 s for the peak and 20 s for the background. Wash out time between individual spots was 10 s. The NIST SRM 612 glass (Jochum et al. 2011) was used as an external standard and the BIR-1G and BHVO-2G glasses (Jochum et al. 2005) as unknowns to monitor precision and accuracy. Five to ten sample measurements were always bracketed by three measurements of NIST SRM 612 glass and two measurements of BIR-1G and BHVO-2G glass. Minor and trace element contents of clinopyroxene and feldspar from the different textural groups are reported (Supplementary data; Sheets 1 and 3). Standard analyses, relative standard deviations (RSD), and GeoReM preferred and reported values are shown in Supplementary data (Sheet 6).

For geochemical analyses, samples were crushed in a steel jaw crusher, before being further pulverized with agate pestle and mortar and processed with a planetary ball mill, to achieve whole-rock powders. Powders were ignited to 1000 °C in for 30 min to determine the loss of ignition and fused to glass beads in Pt crucibles at 1150 °C after mixing with $Li_2B_4O_7/LiBO_2$ flux. Whole-rock major-element concentrations were determined in the glass beads by X-ray fluorescence spectroscopy (XRF) on a Panalytical AxiosMax at VU Amsterdam, following the analytical procedures outlined in Klaver et al. (2016). Trace element concentrations of bulk sample powders were determined using a Thermo Scientific iCAPQc quadrupole ICP-MS instrument at the Scripps Isotope Geochemistry Laboratory, Scripps Institution of Oceanography (USA), following the procedure outlined in Day et al. (2014). Measured values of samples and standards for bulk rock analyses are reported in Supplementary data (Sheet 7).

Results

Petrography

The studied samples are crystal-rich (40–55 vol. %) and display, to varying degrees, a porphyritic texture with discrete phenocrysts set in a microlite-laden groundmass. Three groups are assigned based on both texture and bulk rock composition: less porphyritic basaltic andesite (LPBA), less porphyritic andesite (LPA) and highly porphyritic andesite–dacite (HPAD). Lava flows from Sterna and the eastern slope of the island constitute the LPBA group. The Agriokastro blocky lava flows form the LPA group. The main part of the island, including discrete lava domes, form the HPAD group. All units have comparable phenocryst assemblages, composed of plagioclase (*pl*), orthopyroxene (*opx*), clinopyroxene (*cpx*), Fe-Ti oxides (Ti-*mt*), and olivine (*ol*) (Fig. S1a). Quartz (*qz*), biotite (*bt*), and scarce amphibole (*amp*) are also present in the most evolved HPAD samples. Lath-shaped *pl* (0.5–1.5 mm) is the most abundant phenocryst phase (20–40 vol.%), with variable zoning patterns, including (a) low amplitude oscillatory zoning (Fig. 2a), (b) normal zoning in inner domains, mantled by sieve-textured *pl* and overgrown by rims of clear *pl*, and (c) fine-scale oscillatory zoning around resorbed calcic cores (Fig, S1b). Pyroxenes (10–18 vol. %) are the next most abundant minerals, with *cpx* predominant over *opx*. The latter occurs as homogeneous crystals up to 0.8 mm, often exhibiting more mafic rims (Fig. S1c, d). Clinopyroxene is mostly present as nearly homogenous crystals, with weak normal zoning (Fig. 2b) and rare reverse rims. A small subset observed in all the examined samples includes *cpx* with mafic inner domains rimmed by more evolved *cpx* (Fig. 2c), with a diffuse and irregular boundary and often patchy zoning. Olivine (up to 1 mm) displays homogeneous cores and more evolved mantles that are rimmed by *opx* (Fig. 2d). Widespread glomerocrysts of $pl \pm cpx \pm opx$ and interstitial groundmass (Fig, 3a), display zoning patterns similar to the phenocrysts. Orthopyroxene-bearing glomerocrysts commonly lack *cpx*, whereas when *cpx* and *pl* are the main glomerocryst constituents, *opx* is often interstitial (Fig. S1d, e, f). A second type of glomerocryst (felsic clots), observed in HPAD, is composed of megacrystic *pl* (3–7 mm; even up to 10 mm) ± resorbed *qz* and partially decomposed *bt* (Fig. S1g). The latter occurs as slightly turbid (Fig. S1h) (HPAD) to pervasively destabilized (LPA, HPAD) crystals, replaced by fine grained $opx + pl + mt \pm ilm$ (Fig. 3b). The decomposed aggregate pseudomorphs are often associated with completely sieved *pl* megacrysts (Fig. 3b). In HPAD megacrysts, inner domains of clear *pl* are also observed within the sieved domains (Fig. 3c), which contain abundant melt channels and *px* inclusions and are often rimmed by clear *pl*. A distinct *cpx* population observed in both LPA and HPAD consists of homogeneous crystals (up to 400 µm) arranged in rosette-shaped structures (Fig. 3d) and containing variable amounts of interstitial glass. Within a single LPBA sample, we report rare schlieren of rhyolite (up to 1 mm thick), composed of quartz and feldspar phenocrysts and pseudomorphs after mafic phases (e.g., *bt*), within a vesicular, microlite-poor glassy groundmass (Fig. S1i). Additionally, two amphibole grains with obvious decomposition features were identified within an HPAD dome sample.

Whole rock compositions

Major elements

Antimilos lavas range from basaltic andesite to low Si-dacite (Fig. 4a), with a calc-alkaline affinity. The observed range generally overlaps with previously reported data (Sonder

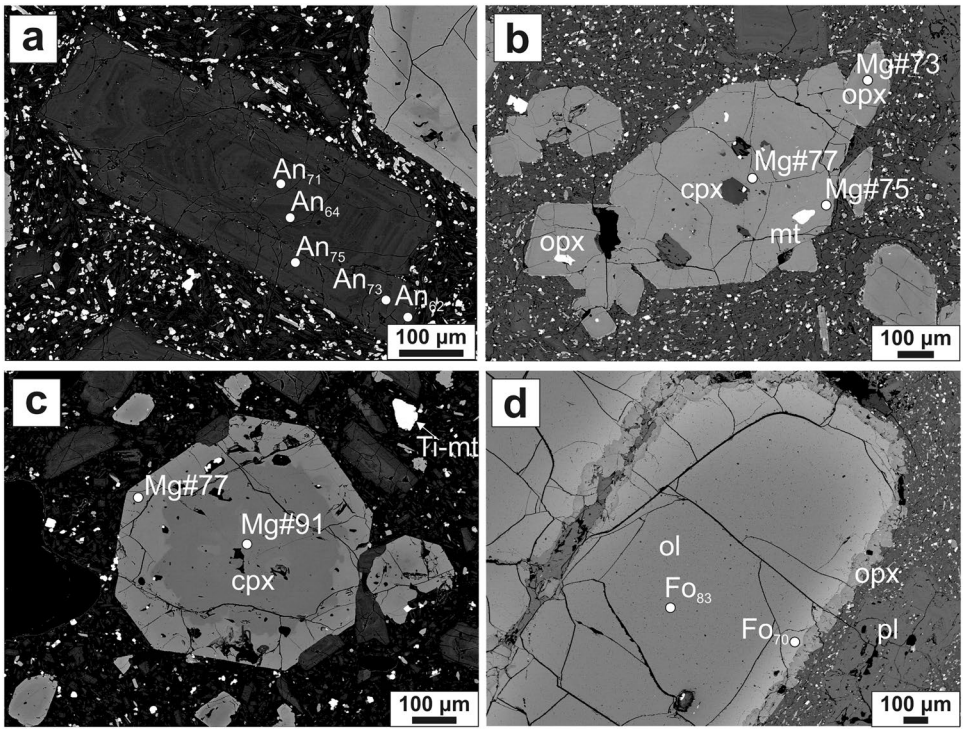


Fig. 2 Representative BSE photomicrographs of phenocrysts from Antimilos lavas. **a** Low-amplitude oscillatory zoning in lath-shaped *pl*. **b** Unzoned *cpx* crystal with touching rims with *opx* micro-phenocrysts. **c** Zoned *cpx* with abrupt boundaries separating a mafic core (BSE dark) and a more evolved (BSE bright) rim. **d** Olivine exhibiting normal zoning and development of an *opx* rim at the contact with the host melt

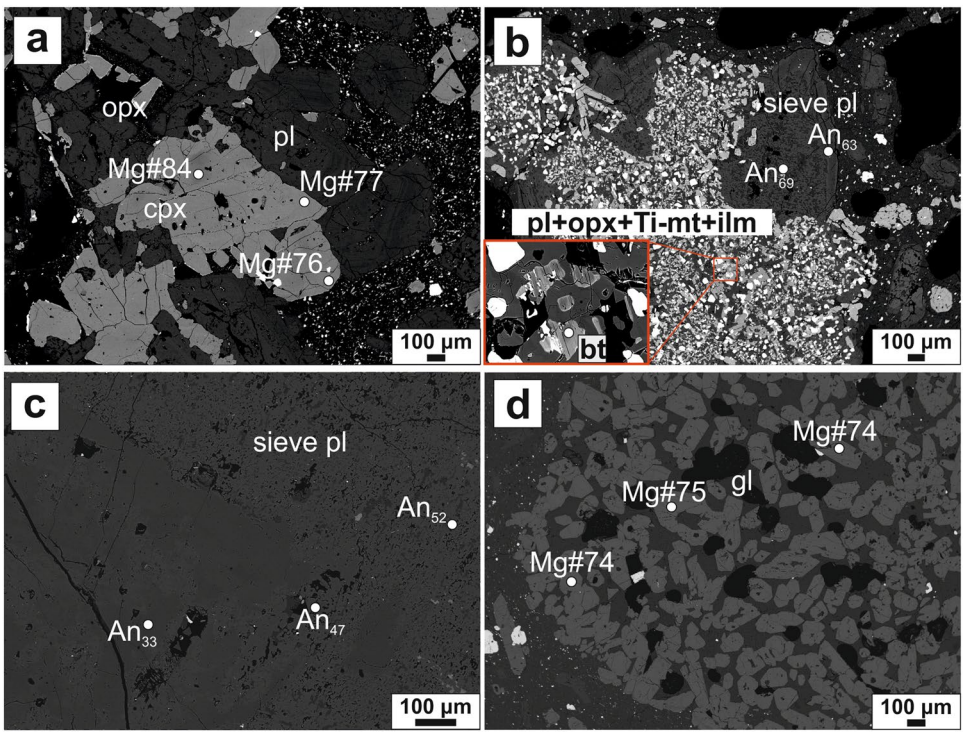


Fig. 3 Representative BSE photomicrographs of crystal clots and disequilibrium textures in Antimilos lavas. **a** Mafic crystal clot composed of *cpx* and *pl* with subordinate interstitial *opx*. **b** Pervasively destabilized felsic clot, now composed of sieve textured *pl* and the fine-grained aggregate of *pl* + *opx* + *Ti-mt* + *ilm* grown at the expense of *bt*. Biotite is rarely preserved within reaction products. **c** Domains of Ab-rich clear *pl*, rimmed by sieve textured *pl* of more calcic composition. **d** Rosette shaped aggregate of *cpx* micro-crysts with interstitial silicic glass, comprising an end-product of *qz* reaction with mafic melt

1924; Marinos 1961; Fytikas et al. 1986), except that we did not find any samples as silica rich as the single rhyolite reported by Fytikas et al. (1986). We do, however, observe mingling with rhyolite schlieren in LPBA containing evolved glass (SiO_2 76.65 wt% and total $Na_2O + K_2O$ 8.28 wt%) that indicates the existence of more felsic components. Major oxide contents are all strongly correlated with silica content forming well defined linear trends (Fig. 4): with increasing SiO_2, we observe decreasing TiO_2, Al_2O_3, MgO, FeOt, MnO, and CaO alongside increasing K_2O and Na_2O.

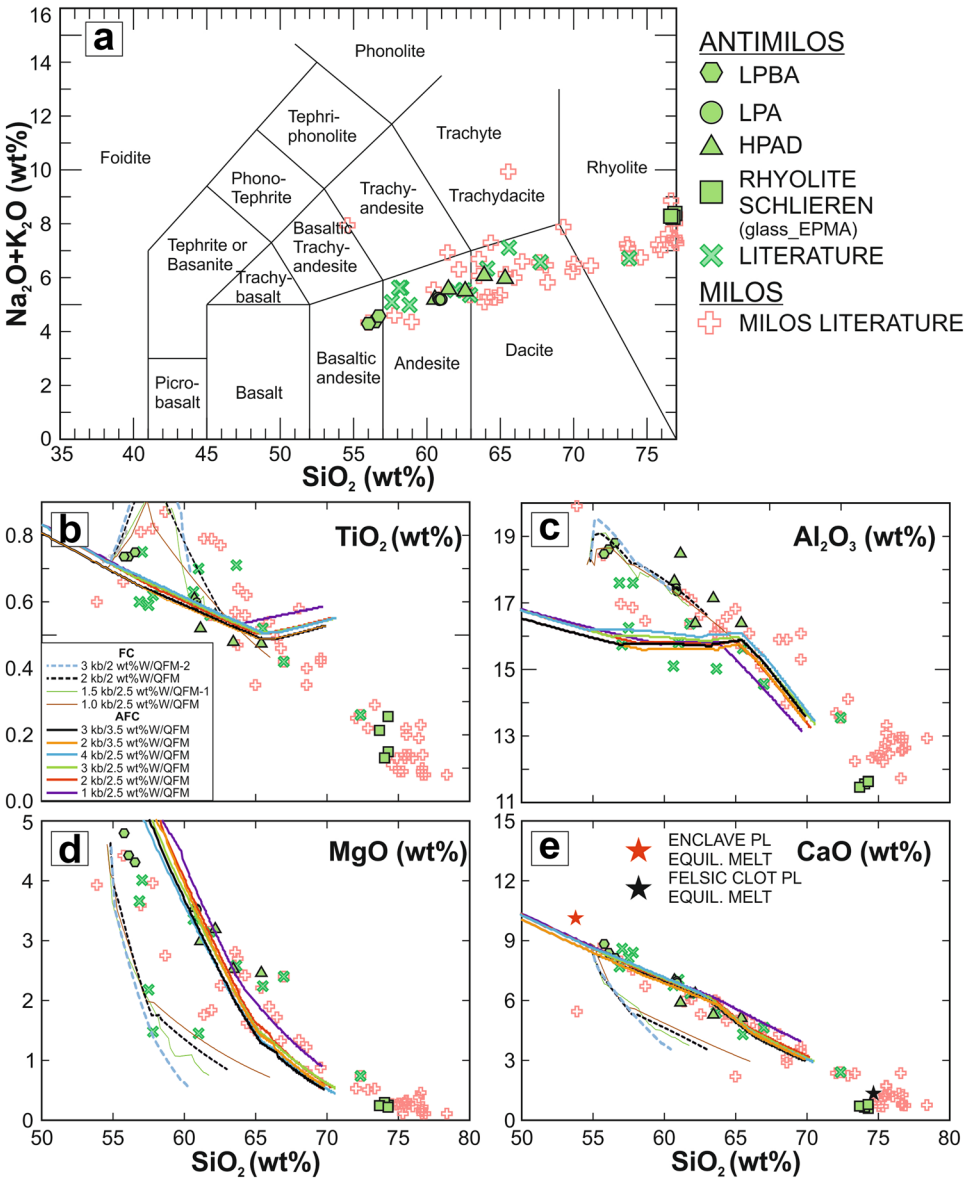


Fig. 4 Whole-rock major-element chemistry of Antimilos samples. **a** Total Alkali-Silica diagram (Le Bas et al. 1986) for classification of Antimilos whole rock samples and groundmass glasses of rhyolite schlieren (EPMA data) **b–e** Harker diagrams of selected major elements against SiO_2. Previous data from Antimilos (Sonder 1924; Marinos 1961; Fytikas et al. 1986;) and Milos (Barton et al. 1983; Fytikas et al. 1986; Zhou et al. 2021) are plotted for comparison. Melts potentially in equilibrium with enclave and felsic clot plagioclase calculated with the method outlined in Scruggs and Putirka (2018) are also shown in **e**. Candidate liquid lines of decent for different fractional crystallization and assimilation plus fractional crystallization scenarios calculated with alphaMELTS are plotted for comparison with the whole rock trends. The better correlations and lower scatter evident in the current Antimilos dataset, is attributed to measurements in a single laboratory with higher precision methods

Trace elements

Compatible transition metals (Cr, Ni, Co, Sc) are relatively low in concentration and negatively correlated with SiO_2, with the highest values observed in LPBA (Fig. S2). Strontium, La, Ce, and Ba (Fig. S2d) follow positive trends versus SiO_2 from LPBA to LPA, but then exhibit inflections and wide ranges among HPAD samples. Incompatible element concentrations increase, as expected, with differentiation, most notably Rb, Th, Nb, and Y (Fig. S2). Trace element ratios like Zr/Hf (Fig. S2g) and the amphibole sensitive Dy/Yb (Davidson et al. 2007; Fig. S2h) both decrease with SiO_2.

N-MORB-normalized trace-element abundance patterns (Sun and McDonough 1989) are typical of subduction-related magmas, with enrichment of large-ion lithophile elements, negative anomalies in Nb and Ti, and positive Pb anomalies (Fig. 5a). Rare earth elements show concave-upwards chondrite-normalized patterns (Fig. 5b) with high LREE/HREE (La/Yb_N ~ 6.7–10). A moderately steep slope is observed from the LREE to the MREE (La/Sm_N ~ 3.4–4.6), coupled with weak negative Eu anomalies (Eu/Eu^* ~ 0.73–0.85); and slight depletion of the MREE relative to the HREE ($Dy/Dy^* = [Dy_N/(La_N^{4/13} \times Yb_N^{9/13})]$ ~ 0.5–0.6; Davidson et al. 2013).

Mineral chemistry

Plagioclase

Antimilos plagioclase ranges from 24 to 94% in the molar anorthite component, bimodally clustering into a low-An (oligoclase-andesine) and a high-An (labradorite-bytownite)

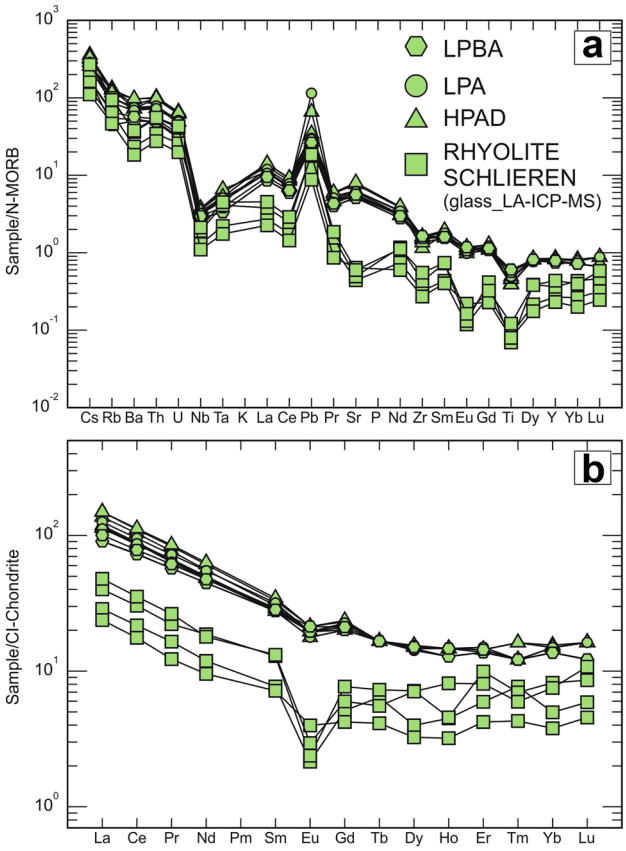


Fig. 5 **a** N-MORB normalized trace element and **b** chondrite normalized REE patterns, for Antimilos whole-rock samples and groundmass glasses of rhyolitic schlieren (LA-ICP-MS) in the basaltic andesite sample AM-11 (LPBA units)

group (Fig. 6a–h). Evolved compositions (An_{24-59}) with relatively low FeO (Fig. 6g-h), Sr, Mg, and elevated Ba (Fig. S3) are commonly encountered in megacrystic and felsic clot plagioclase in HPAD units, as well as within the rhyolite schlieren plagioclase (An_{27-47}), and are substantially distinct from the main phenocryst population. The latter stands out to more calcic compositions and elevated FeO (Fig, 6 a-b), Sr, Mg contents (Fig. S3), displaying remarkable overlap with glomerocryst plagioclase (Fig. 6c), with the only notable differences the increased abundance of primitive (An_{85-94}) cores with negative correlation between X_{An}-FeO (Fig. 6d) in the glomerocrysts. Core–rim variations in the different textural types are in the range of 60–80% molar anorthite, with more evolved compositions restricted in interstitial *pl* and rims. Plagioclase cores and rims from a quenched enclave follow the same compositional trends as the glomerocrysts (Fig. 6d). Both the low- and high-An compositional clusters are observed in sieve-textured megacrysts (Fig. 6e, f), with clear internal domains being sodic (An_{27-41}; low FeO, Mg, and Sr), and sieve domains (An_{45-82}) showing a more pronounced positive correlation between X_{An} and both FeO ($R^2 = 0.31$; Fig. 6f) and Mg ($R^2 = 0.1$; Fig. S3a). Clear *pl* (An_{56-77}; 0.29–0.95 wt% FeO) is often rimming the sieve-textured domains, sharing similar compositions (Fig. 6a–f) with phenocryst and glomerocryst rims.

Orthopyroxene

Low-Ca pyroxene phenocrysts have Mg# [Mg/(Mg + Fe) × 100] varying from 63 to 79; more magnesian *opx* is rare and observed only as inclusions in *cpx* (Fig. 7a, b). A negative correlation is observed between Mg# and MnO in most analyses, although the most evolved points (Mg# ≤ 70) show a reversal in the trend (Fig. 7b). This break in slope is also present in enclave *opx*, but at Mg# = 75. In glomerocrysts *opx* is slightly more magnesian (Mg# 67–82) with relatively high MnO. A compositionally narrow range (Mg# 72–77) is observed in *opx* rimming *ol* crystals. The *opx* (Mg# 67–76) in *bt* breakdown products is notably enriched in Al_2O_3.

Clinopyroxene

Antimilos clinopyroxenes range from diopside to augite (Fig. 7a) but exhibit distinct compositional features in respect to their textural association. The main *cpx* population is found in relatively homogenous crystals (Mg# 75–83), overlapping with *cpx* in clots, and evolved rims within a small subset of zoned *cpx*. The latter exhibit mafic core compositions (Mg# 86–93) with a positive correlation between Mg# and Al_2O_3 (1.52–4.66 wt%; $R^2 = 0.51$), being prominent in *cpx* with Mg# > 85 (Fig. 7c). Both groups hold substantially distinct trace element budgets; the mafic cores show lower Mn, V (Fig. 7d), Sc, Y (< 20 µg/g) (Fig. 7e), Ti and Zr, but higher Cr (> 200 µg/g) and Sr, compared to evolved *cpx*, which additionally display higher absolute REE abundances (Fig. 8a-f) and negative Eu anomalies (Eu/Eu* = 0.44–0.80). Clinopyroxenes in rosette-shaped aggregates are characterized by the lowest (Fig. 7c) Al_2O_3 for a given Mg#, and the highest Mn values among all Antimilos *cpx*. Chondrite normalized REE patterns (Fig. 8d, f) exhibit a 'seagull shape' with variable REE enrichment (ΣREE ~ 78–285 µg/g) and negative Eu anomalies (Eu/Eu* = 0.52–0.77).

Olivine

Olivine (Fo_{70-85}) exhibits normal zoning with decreasing forsterite component towards the rims, where an *opx* overgrowth is commonly observed in contact with the BSE bright *ol* rims. Manganese content (0.15–0.65 wt% MnO) is also zoned, increasing towards the rims and is, hence, anticorrelated with forsterite.

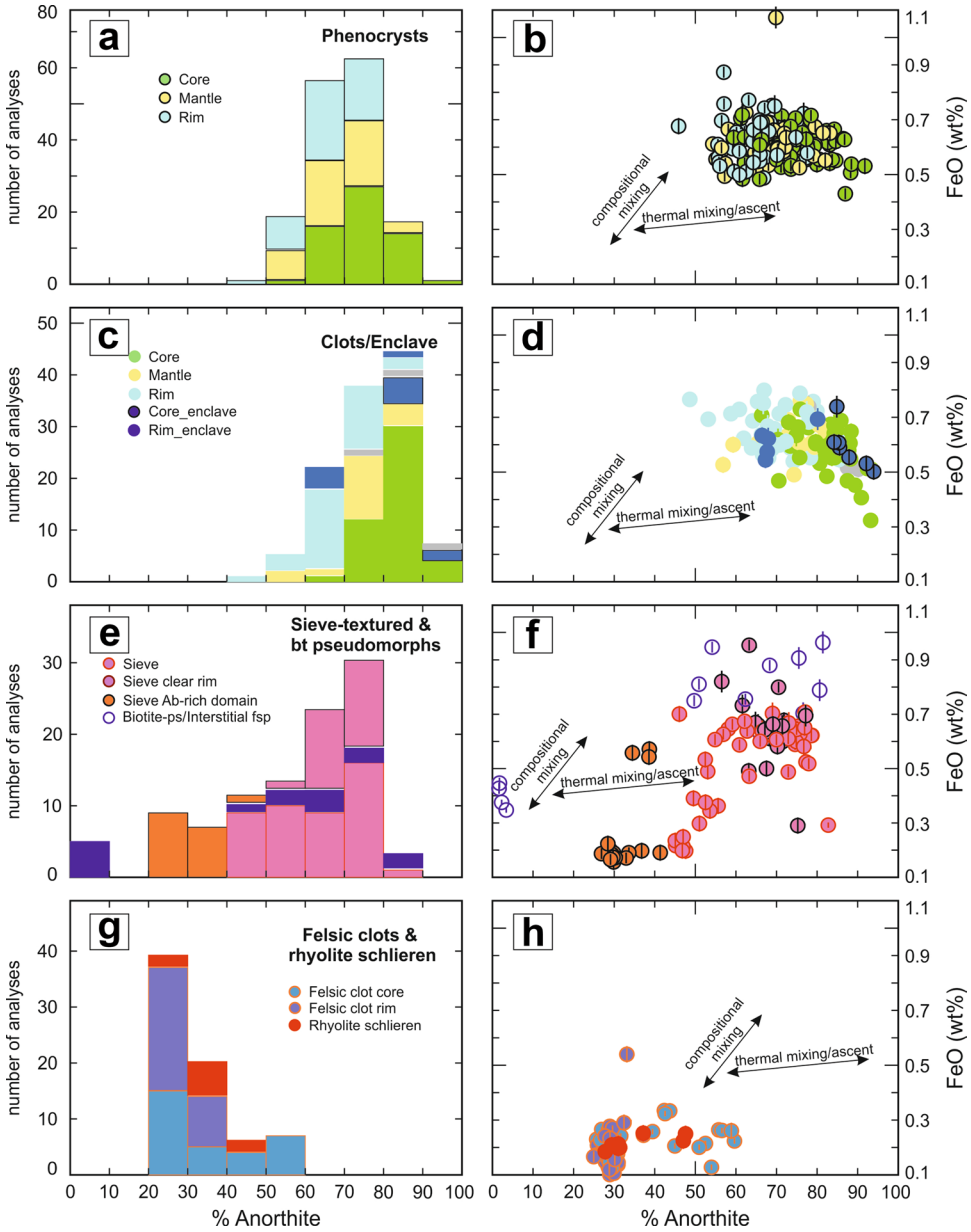


Fig. 6 Comparative mineral chemistry of Antimilos plagioclase found in different textural associations. **a**, **c**, **e**, **g** Histograms showing the distribution of % molar Anorthite component in *pl* with respect to the mode of occurrence and inter-crystal stratigraphy. **b**, **d**, **f**, **h** Variation of % molar anorthite component with FeO (EPMA) in *pl* in respect to the mode of occurrence and inter-crystal stratigraphy. Vectors indicating the effects of compositional and thermal mixing/ascent in FeO content in *pl* are drawn after Ruprecht and Wörner (2007). Error bars represent 1-sigma errors on the EPMA FeO measurement

Biotite and amphibole

Biotite is found in Agriokastro andesites samples only as scarce relics within pseudomorphs, characterized by Mg# 66–70, almost constant $^{IV}Al/^{IV}(Al+Si)$ ratio (0.30–0.31) (Fig. S4a), and uniformly high TiO_2 contents (7.0–8.6 wt%). In porphyritic andesite–dacite units, *bt* displays a larger range (Mg# 51–78 and $^{IV}Al/^{IV}(Al+Si) \sim 0.24–0.34$), likely reflecting the presence of different textural groups. There is also a large range in TiO_2 (3.68–10.3 wt%), which shows negative correlations with the other octahedral site cations (Fig. S4a). Scarce amphibole is observed in a single HPAD dome (sample AM-2). It has high Al content (1.77–2.08 a.p.f.u), a limited range of TiO_2 content (1.78–2.48 wt%) and Mg# 67–72. It is classified as Mg-hastingsite to Ti–rich Mg-hastingsite (Fig. S4b) in the nomenclature of Hawthorne et al. (2012).

Discussion

Estimation of *P–T*-H_2O record of the crystal cargo

The intensive parameters of magma storage within the Antimilos sub-volcanic plumbing system were estimated (Fig. S5a-d) utilizing the software package Thermobar (v.1.0.12, Wieser et al. 2022), using appropriate single-phase and mineral-melt equations and equilibrium filters. Additional

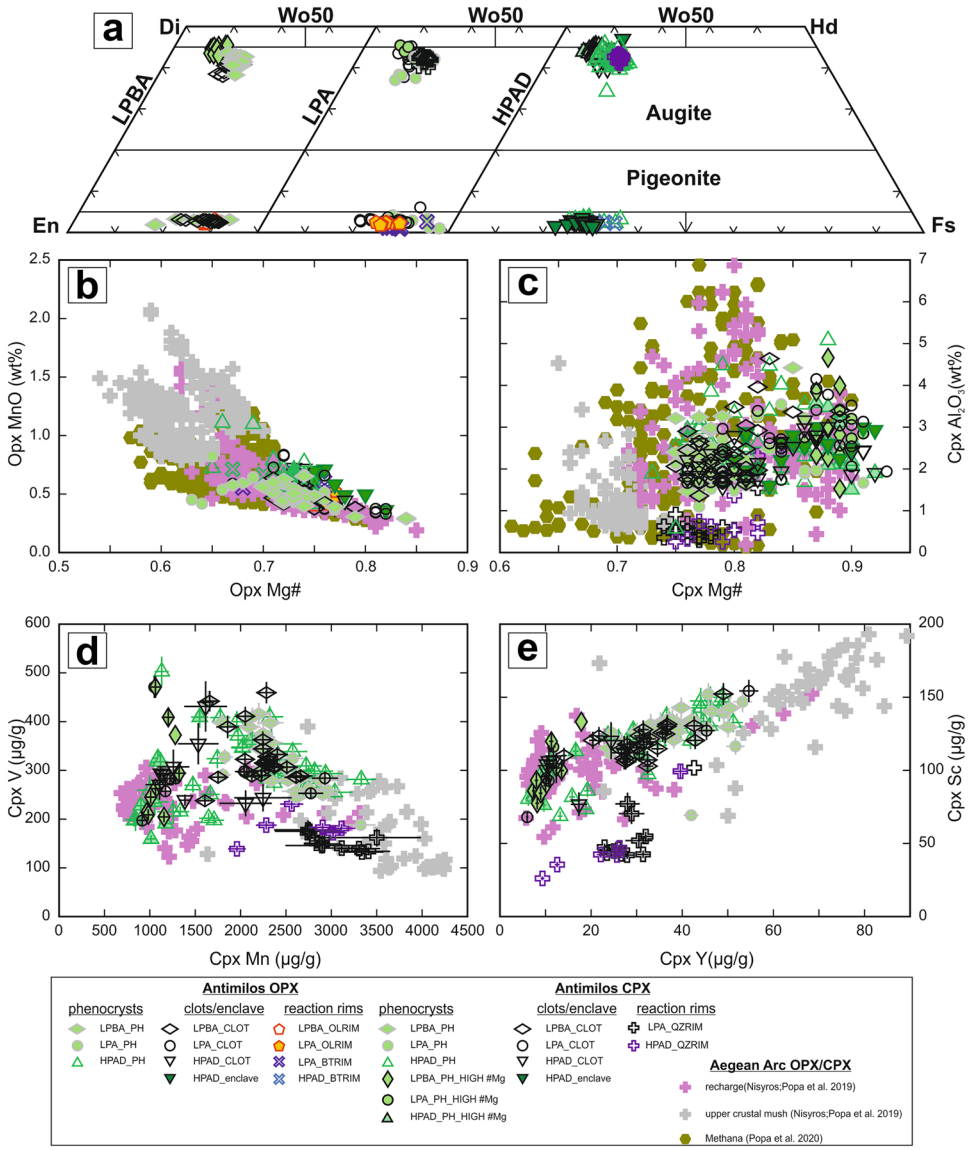


Fig. 7 Major and trace element geochemistry of Antimilos pyroxenes with respect to their mode of occurrence. **a** Quadrilateral classification of distinct *opx* and *cpx* groups. **b** Variation of MnO vs. Mg# for the studied *opx*. **c** Variation of Al_2O_3 with Mg# showing two distinct trends in *cpx* from phenocrysts and clots and a distinct low Al_2O_3 cluster in *cpx* formed because of *qz* dissolution. **d** Variation of Mn versus V and **e** Y versus Sc for the different *cpx* groups. Available pyroxene data from Nysiros and Methana (Popa et al. 2019; 2020) are plotted on top of Antimilos to discriminate between recharge and upper crustal mush derived pyroxenes. Error bars represent 1-sigma errors of the LA-ICP MS measurements

filtering was applied in the produced P–T dataset, to exclude mineral composition corresponding to points where we observed clear disequilibrium textures (e.g., reaction-related *cpx* and reverse-zoned rims) and may return unrealistic conditions estimates due to insufficient mineral-melt matching.

Considering that equilibrium *cpx*-liquid pairs are rarely preserved, we applied the machine-learning based *cpx*-only thermobarometer of Jorgenson et al. (2022) to get a tentative estimate of storage conditions. Despite the lack of good performance of the above method, especially in the pressure estimates (Wiesser et al. 2023), it is suitable for our case, since it can be applied to the different *cpx* textural groups without prior knowledge of equilibrium melt compositions (e.g., H_2O melt). The *cpx*-only temperature estimates (Fig. S5b) of the phenocryst (low Mg#) and mafic clot populations overlap (985–1150 ± 51 °C) returning similar median pressures (200 ± 320 MPa) even though a much wider range of pressures is recorded in clot *cpx* (0–500 ± 320 MPa) compared to phenocrysts (0–220 ± 320 MPa). High-Mg# clinopyroxene cores return slightly hotter (1020–1140; mean 1071 ± 51 °C) and higher pressure (200–480; mean 218 ± 320 MPa) estimates.

Orthopyroxene-liquid thermobarometry (Fig. S5c-d) was applied to *opx* analyses that passed a test for equilibrium with their whole-rock hosts, melt inclusions and groundmass glasses using the P-T independent range of $K_D^{Opx-Liq} = 0.29 \pm 0.06$ from Putirka (2008). Liquids found in equilibrium with *opx* usually contain 67.7–72.3 wt% SiO_2, supporting the late appearance of *opx*. For these cases, we estimated P from the H_2O-independent global calibration of Putirka 2008 followed by a range of saturation T estimates (Eq. 28b; Putirka 2008) allowing H_2O contents in the melt between 1 and 6 wt%. Estimated temperatures span a narrow range (963–984 °C for 2 wt% H_2O in melt; 950–974 °C for

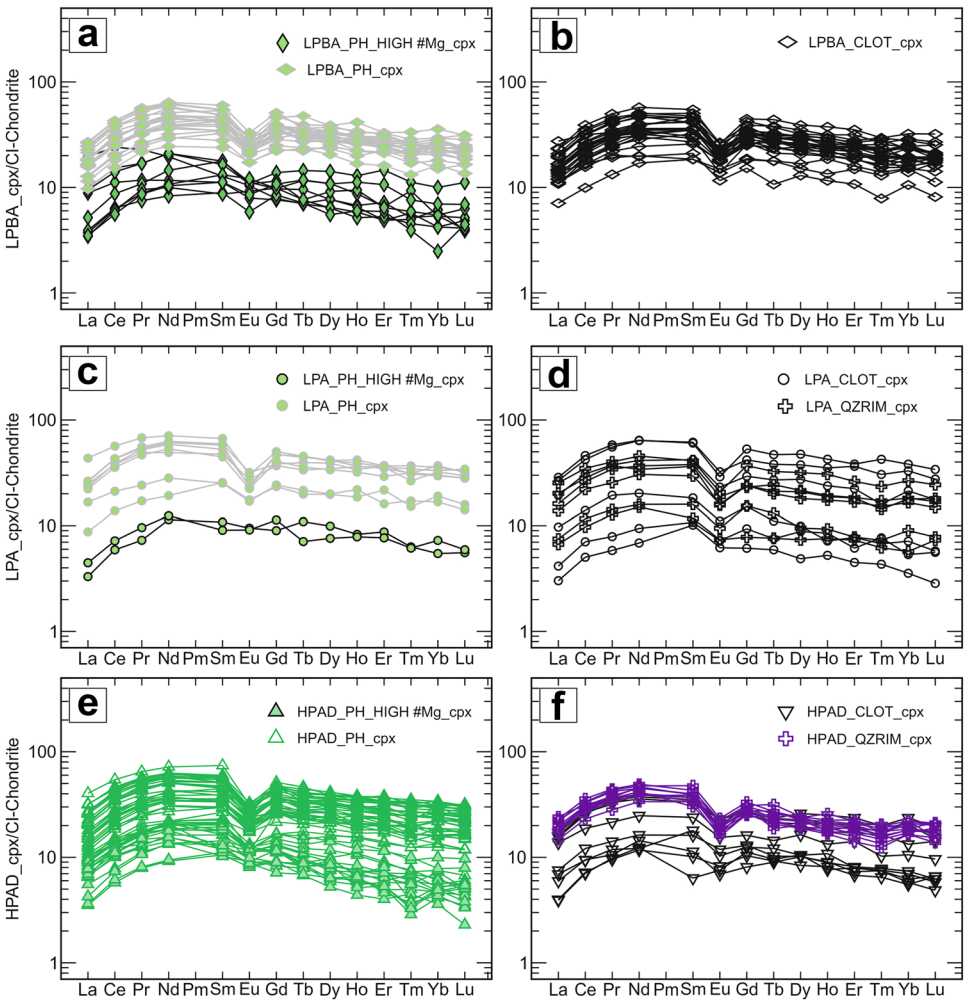


Fig. 8 Chondrite normalized (McDonough and Sun 1995) REE patterns for the texturally distinct *cpx* groups found in Antimilos units. **a**, **c**, **e** phenocrysts and high Mg# cores **b**, **d**, **f** crystal clots and quartz replacement rims for LPBA, LPA and HPAD, respectively

3 wt% H_2O in melt) while pressures are scattered from 62 to 370 ± 260 MPa (median 130 ± 260 MPa).

Another set of constraints on magmatic temperatures and H_2O contents of the melt were obtained using the *pl*-liquid thermometer (Eq. 23 of Putirka 2008) and the hygrometer of Waters and Lange (2015), matching plagioclase compositions with Antimilos bulk rocks and groundmass glasses from the rhyolite schlieren. These two equations were iteratively solved at fixed pressure (100–200 MPa), considering only *pl*-liquid pairs that pass the Ab-An exchange equilibrium test of Putirka (2008) and excluding sieve-textured plagioclase. Iterating T-H_2O is particularly useful considering the high temperature-sensitivity of the hygrometers (Waters and Lange 2015), and the fact that an independent temperature estimate may result from phases (e.g., *cpx*, *opx*) that are not recording the same magmatic histories (Wiesser et al. 2022), and additionally the negligible ΔT and ΔH_2O after 20 iterations, indicate that the iterations converge. Plagioclase-liquid temperatures (767–1100 ± 43 °C) and melt H_2O contents (2.02–4.87 ± 0.35 wt% H_2O) are strongly bimodal (Fig. S5a), indicating that plagioclase formed in at least two distinct magmatic environments. Temperature estimates for the low-T (767–988 ± 43 °C; average 802 °C) and high-H_2O (3.14–4.87 ± 0.35 wt%) group correspond to plagioclase found in felsic clots and rhyolite schlieren. By contrast, the high-T (911–1100 ± 43 °C) and low-H_2O (2.02–3.14 ± 0.35 wt%) plagioclase group (Fig. S5a) corresponds texturally with phenocryst, mafic clot, and enclave *pl*.

Crystal cargo record of three distinct magmatic environments

Textural and compositional evidence from the crystal cargo of Antimilos lavas suggests that the erupted units record the presence of distinct compositional and thermal environments within the Antimilos plumbing system. In the following sections, we attempt to assess the mineral assemblages which formed under different equilibrium conditions, to provide insights about the magmatic components residing in the plumbing system and to explain how these reservoirs contribute to the erupted record. A compilation of equilibrium

assemblages, conditions and crystal cargo provenance in the erupted units is summarized in Table 1.

An evolved upper crustal reservoir record in felsic antecrysts

Despite the limited compositional range of the bulk rocks, the crystal cargo provides strong evidence for the presence of even more evolved components ($SiO_2 > 70$ wt%) within the Antimilos plumbing system. This more evolved component may be the felsic melt inferred to be assimilated into the fractionating assemblage to explain the detailed compositional variation in the sampled suite. Plagioclase in the HPAD with compositions as evolved as An_{30} might be taken to suggest continuing fractionation of the Antimilos liquid, eventually reaching strongly evolved compositions. However, this simple model is inconsistent with the paucity of *pl* compositions between An_{40} and An_{60} and with the zoning patterns in *pl* phenocrysts and glomerocrysts, with rim compositions, commonly ranging around $68 \pm 8\%$ molar anorthite. Likewise, the modest Eu anomalies (Eu/Eu* ~ 0.73–0.77) and high Sr contents in HPAD bulk rocks do not favor extended fractional crystallization to reach a felsic endmember. As an alternative, we examine the derivation of low-An *pl* megacrysts from a pre-existing evolved component, resident in the upper crust and rejuvenated by the parental magmas of HPAD. The observed compositional overlap in An, Ba and Sr between the megacrysts and felsic clots in HPAD with *pl* from the rhyolite filaments in LPBA (Fig. 6g, Fig. S3) suggests derivation of these three groups of *pl* crystals from similar melts. Calculated $D_{Sr} \sim 6.54$ and $D_{Ba} \sim 1.6$ from *pl*-glass pairs in the rhyolite filaments and the low Eu/Eu* (0.23–0.72) of the glasses are consistent with melts that experienced significant *pl* fractionation and could therefore crystallize sodic *pl*. Additionally, the empirical model of Scruggs and Putirka (2018, Eq. 2 and 3) yield $SiO_{2melt} = 74.6 \pm 3.9$ wt% and $CaO_{melt} = 1.3 \pm 1.2$ wt% for liquids in equilibrium with HPAD megacryst and felsic clot *pl*, which resembles the filament glasses and is clearly distinct from all Antimilos bulk rocks. The presence of disseminated resorbed quartz and biotite within the HPAD and their common association with An_{30} *pl* in crystal clots further indicates assimilation of material that reached much lower temperatures than those recorded by the Antimilos lava bulk

Table 1 Summary of the different assemblages, mineral compositions and conditions recorded in Antimilos lavas

Assemblage	Pl + cpx + opx + mt ± ol	Cpx ± amp, ± ol, ± pl	Pl + bt + qz	Rection rims
Occurrence	Phenocrysts and crystal clots	Antecrystic cores	Antecrysts and antecrystic clots	Sieve textures Pseudomorphs Coronas
Plagioclase	An 60-80 0.5–0.8 wt% FeO No corel. An-FeO	An > 85 0.3–0.6 wt% FeO Negative An-FeO	An 24-59 < 0.3 wt% FeO Positive An-FeO	Sieve-textured An 45-82 0.2–0.9 wt% FeO Replacing bt An 50-81
Clinopyroxene	Mg# 75–83 Low Sr/Y Eu/Eu* Negative Intermediate Mn	Mg# 86–93 High Sr/Y Eu/Eu* absent-slightly negative Low Mn	Absent	Replacing qz Mg# 82–74 High ^{IV}Si Eu/Eu* negative High Mn
Orthopyroxene	Mg# 63–79 < 0.6 wt% MnO	Absent	Absent	Rimming ol Mg# 72–77 Replacing bt Mg# 67–82
Olivine	Trace, not stable	Mg#70–85	Absent	Absent
Biotite	Absent	Absent	Mg# 51–56 0.2 a.p.f.u. Ti	Mg# 67–79 0.2–0.6 a.p.f.u Ti
Amphibole	Absent	Relic High Al Inferred from cpx trace elements	Absent	Absent
Temperature range	911–1100 ℃ (pl-liq) 985–1150 ℃ (cpx) 910–1000 ℃ (opx-liq)	1020–1140 °C (cpx)	< 850 ℃ (bt exp.) 775–850 ℃ (pl-liq) < 800 ℃ (qz exp.)	Not Estimated
Storage region	Upper(-Mid) Crust	Mid(-Lower) crust	Upper crust	Upper crust
H_2O melt	~ 3 wt% (pl-liq)	Higher than ME-B	> 4 wt% (pl-liq)	Not estimated
Provenance in erupted units	Crystallization during upper-crustal storage	Cargo from mafic recharge	Rejuvenated silicic mush	Recharge-mush interaction

rocks and phenocrysts. Specifically, at middle to upper crustal pressures, *bt* forms below 850–870 °C (First et al. 2021; Marxer et al. 2022) and co-existing *qz* suggests T < 800° (Muir et al. 2014). Antimilos plagioclase-liquid T estimates are consistent with the above range, especially in the low-T end. We label this setting, characterized by low temperatures and evolved compositions, as magmatic environment-A (ME-A) to contrast it with evidence for other crystallization environments sampled beneath Antimilos. Felsic antecrysts in Antimilos are likely derived from such an environment (e.g., rhyolitic melt pockets, mush, older intrusion), as their formation along a common liquid line of descent is inconsistent with observed compositions.

Phenocryst recording hot, low-pressure storage of mafic recharge melts

Petrographic observations and geochemical analyses suggest that the three units of Antimilos lava share a similar phenocryst assemblage (*pl* + *cpx* + *opx* + *mt* ± *ol*). Plagioclase rims (An_{50-80}) of phenocrysts and glomerocrysts overlap in composition with the clear rims of sieve-textured *pl* (Fig. 6a-f), especially in the range An_{60-80}. Poor correlation between anorthite and FeO contents (Fig. 6b, d) suggests that the observed variations in plagioclase composition result from processes like thermal mixing and/or during magma ascent (e.g., Ruprecht & Wörner 2007). Geochemical fingerprints of *pl* crystallization are recorded in the low-Mg# (75–83) population of *cpx* phenocrysts and *cpx* in crystal clots, such as decreasing Al_2O_3 with Mg# (Fig. 7c), low Sr/Y and negative Eu anomalies (Fig. 8). The negative correlation between V and Mn in *cpx* (Fig. 7d), suggests co-crystallization of *cpx* with a V-sequestering phase (e.g., Ti-*mt*). Two-pyroxene andesites that may lack olivine (given the presence of *opx* rims) are commonly reported from primitive arc basalt crystallization experiments, at low pressures (100–400 MPa) and temperatures between 940 and 1050 °C (Sisson & Grove 1993; Pichavant et al. 2002; Grove et al. 2003; Blatter et al. 2013; Andujar et al. 2015; Marxer et al. 2022).

Our estimates indicate that plagioclase dominates the fractionation sequence co-existing with low Mg# clinopyroxene in the high temperatures, whereas *opx* joins late in the sequence (T < 1000 °C) and becomes the main mafic phase in Antimilos lavas down to 950–930 °C. We refer to this as magmatic environment-B (ME-B), corresponding to the thermochemical conditions established in the upper crustal reservoir, following the arrival and storage of mafic recharge, prior to eruption. The recharge affinity of the phenocryst cargo in Antimilos is better demonstrated when comparing pyroxene chemistry with other volcanic centers in the Aegean. In Mg#-MnO space, Antimilos *opx* phenocrysts compositionally overlap with mafic recharge crystals from Nisyros and Methana (Popa et al. 2019; 2020b), showing good agreement with experimental *opx* crystallized from basaltic magmas at upper crustal pressures (200 MPa; Andujar et al. 2015). The abundance of *pl* and late *opx* additionally suggests low pressures for this setting (Blundy and Cashman 2008). We argue that phenocryst crystallization in Antimilos erupted lavas, took place during the storage of the recharge magmas and their interaction with felsic components residing in the evolved reservoir. The similarity between plagioclase phenocryst rims and clear *pl* rimming sieve-textured crystals in LPA and HPAD supports this interpretation.

Deep-hydrous fractionation record in mafic antecrysts

Clinopyroxene cores with high Mg# (86–93) and evidence of resorption form a small but important population within Antimilos lavas. A negative correlation observed between Al and Mg# in *cpx* (Fig. 7c) is indicative of crystallization of *cpx* without co-precipitating plagioclase (e.g., Müntener et al. 2001; Pichavant and Macdonald 2007; Villiger et al. 2007; Nandedkar et al. 2014). Such trends are commonly observed in *cpx* found in arc cumulates (Arculus and Wills 1980; Dessimoz et al. 2012; Stamper et al. 2014; Klaver et al. 2017; Higgins et al. 2022). Furthermore, the overall enrichment in diopside component coupled with the more pronounced Tschermak substitution in high Mg# *cpx* (Fig. 9a) may suggest a hotter or more hydrous crystallization environment (Blundy & Cashman 2008; Putirka 2008; Mollo et al. 2018; Petrone et al. 2022). Negligible Eu anomalies in the high-Mg# *cpx* also indicate delayed *pl* crystallization, whereas low REE and Y (< 20 µg/g) coupled with higher Sr (> 40 µg/g) than in the low-Mg# *cpx* phenocrysts are consistent with derivation from a deeper region of the plumbing system, as D_Y and D_{REE} tend to be lower at higher pressures, while the opposite behavior is observed for D_{Sr} (Bédard 2014). Near-vertical trends in Y versus Sr/Y (Fig. 9b) and Dy versus Dy/Yb (Fig. 9c) cannot be explained by *cpx* fractionation alone, but instead require the involvement of *amp*, as the latter has greater influence on Dy/Yb (Davidson et al. 2007, 2013). The above trends are in excellent agreement with *cpx* coexisting with *amp* in various arc cumulate suites (Smith 2014; Cooper et al. 2016; Klaver et al. 2017; Wang et al. 2019). The presence of Fo_{85} olivine suggests derivation from melts with > 6.5 wt% MgO, higher even than the most primitive LPBA, supporting an antecrystic origin for the olivine, and they may in fact even represent equilibrated cores in a bulk of more primitive compositions (Winslow et al. (2020). In a similar manner, *pl* core compositions with $X_{An} > 85$, found in phenocrysts but mainly in crystal clot *pl*, may reflect formation in an earlier stage under more H_2O-rich conditions compared to the main phenocryst population (An_{60-80}).

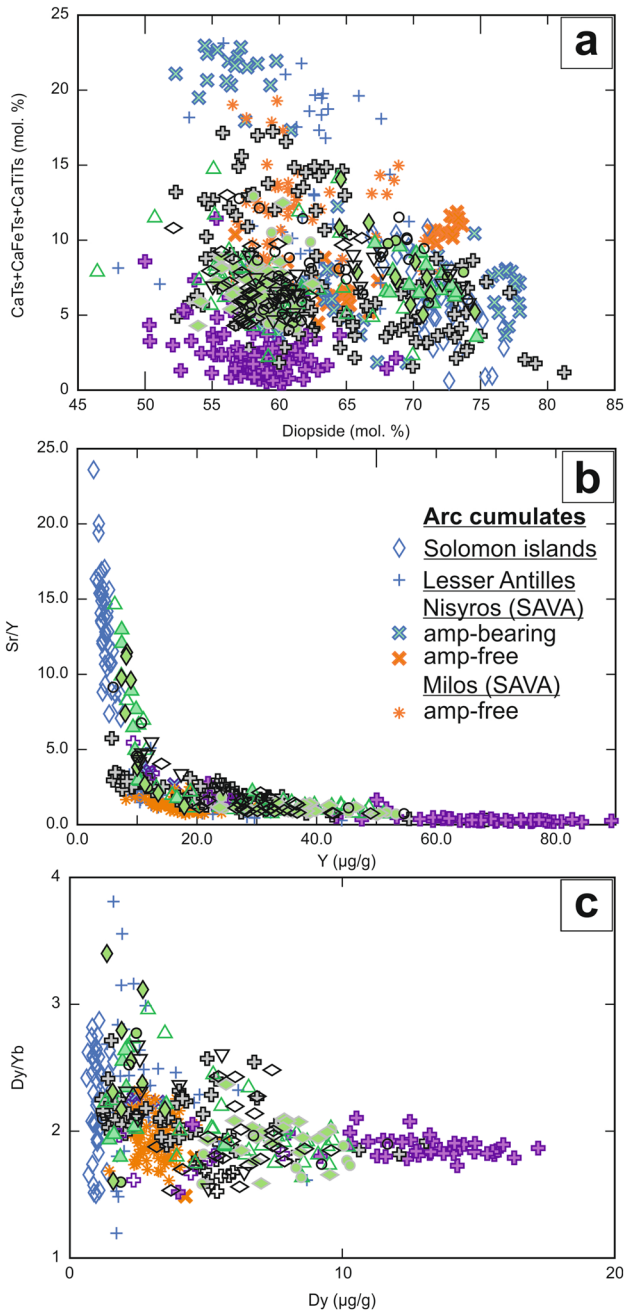


Fig. 9 Variation of **a** molar Diopside versus Tschermak component, **b** Y versus Sr/Y and **c** Dy versus Dy/Yb ratios for *cpx* phenocrysts, antecrystic cores and glomerocrystic aggregates from Antimilos lavas. Clinopyroxene data from arc cumulates from Solomon Islands (Smith 2014), Lesser Antilles (Cooper et al. 2016) and Nisyros (Klaver et al. 2017), Milos (Xydous et al. 2022) are plotted from comparison. High Mg# *cpx* cores from Antimilos closely overlap and follow similar trends with the literature data, indicating a deeper and more hydrous crystallization environment, compared to the one that the phenocrysts grew, and a possible influence of amphibole that may explain the low Y content and variable Dy/Yb ratio. Symbols for Antimilos clinopyroxenes are the same as Fig. 7, symbols for arc cumulates are given by location, except Nisyros and Milos cumulates which were grouped regarding the presence of amphibole

The mafic antecrysts appear to record conditions quite distinct from those inferred for the crystallization of the phenocryst assemblage, and hence it is likely that these antecrysts did not grow in the post-recharge mixed crystallization setting that we label ME-B. Instead, they record a third magmatic environment, here called ME-C, with its distinct geochemical fingerprint lying in the evidence of amphibole involvement in the magmatic processes. This setting is overall characterized by more mafic compositions, and more hydrous and deeper conditions than ME-B, likely representing a mid-lower crustal mush zone from where recharge magmas may originate or scavenge antecrysts from.

Records of the mixing processes

Multiple magmatic environments contributed to the crystal cargo of the erupted units in Antimilos, as indicated by distinct characteristics of phenocrysts, cognate clots, felsic and mafic antecrysts. The processes whereby material from these distinct environments were combined to yield the observed rocks is furthermore preserved by various disequilibrium textures present in the antecrysts. Specifically, we find evidence that the felsic component assimilated from cold ME-A into the evolving ME-B magma carried crystals of *pl*, *qz*, and *bt*, a process known as 'petrological cannibalism' (Cashman and Blundy 2013). Likewise, ascending recharges of hot mafic magma brought material (e.g., *cpx* antecrysts) from ME-C to ME-B.

A widespread population of *pl* in LPA and HPAD exhibits clear internal An_{25-59} domains rimmed by more calcic (An_{45-83}) sieve-textured *pl*. Anorthite-FeO systematics reveal that the clear internal domains, low in both An and FeO, are compositionally equivalent to *pl* found in felsic clots. The positive correlation between X_{An} and FeO (Fig. S6a, b) in the sieve-textured *pl* domains can be interpreted as a record of compositional mixing (Ginibre and Wörner 2007; Ruprecht and Wörner 2007; Shcherbakov et al. 2011). While it is still unclear if sieve texture results from dissolution–reprecipitation reactions between sodic *pl* and mafic melt or from rapid growth due to strong undercooling, the assimilation of a rhyolitic magma containing evolved plagioclase into an evolving mafic melt could trigger development of sieve *pl* by either mechanism (Hibbard 1981; Lofgren and Norris 1981; Anderson 1984; Wark and Watson 1993; Nakamura and Shimakita 1998; Streck 2008).

Clinopyroxene replacing quartz offers additional evidence for assimilation of the latter into a hot mafic melt (Sato 1975), with Antimilos *cpx* rosettes representing end-products of this process (Fig. 3d). The very high Si content (1.96–1.99 a.p.f.u) and low aluminum in rosette *cpx* indicates high silica activity adjacent to dissolving *qz* (Sato 1975; Har and Rusu 2000). Interstitial high-silica glass (SiO_2 varies from 76.2 to 77.0 wt%) with alkali enrichment

($Na_2O + K_2O$ varies from 7.97 to 8.14 wt%) is typical of silicate glasses in *qz*-dissolution coronae (Sato 1975). The complete dissolution of quartz in favor of euhedral, radial *cpx* suggests prolonged reaction times (Luhr et al. 1995) in the presence of hot, crystal-poor mafic melt (Donaldson 1985).

Further textural evidence suggesting assimilation of cold resident magma and/or crystals into the fractionating system is found in reaction rims around biotite (Feeley and Sharp 1996). In this case, experimental evidence suggests an upper bound on reaction time (< 1 year for 60 µm rim growth at T ~ 800–850 °C; Nakamura 1995) for the HPAD clots, in which *bt* is only partly resorbed (Fig. S1g, h; Fig. S4c, d). By contrast, complete replacement of *bt* by pseudomorphs composed of $opx + pl + mt \pm ilm$ (Fig. 3b) in LPA and dispersed phenocrysts in HPAD (Fig. S6c–f) may suggest longer timescales or higher T. While similar rims lacking *mt* have been observed in Cap de Fer (Algeria) dacites and interpreted as evidence of decompression-induced resorption (Fougnot et al. 1996), those magmas lacked evidence of magma mixing or reheating. In combination with reheating-induced disequilibrium textures affecting other minerals (*pl*, *qz*) and the presence of mafic enclaves, we favor increasing T rather than decreasing P as a mechanism of *bt* destabilization at Antimilos. The elevated TiO_2 contents in scarce *bt* relics within pseudomorphs compared to slightly reacted *bt* in clots (Fig. S4a) are also consistent with heating, as Ti substitution in the octahedral site of *bt* is favored at high T (e.g., Robert 1976; Henry et al. 2005).

The textural and chemical evidence of mafic antecrysts also suggests the transport of crystals from ME-C to ME-B. High-Mg# *cpx* with a distinct amphibole geochemical signature is found in cores, separated by a resorption interface, from *cpx* rims that crystallized from more evolved melts affected by simultaneous *pl* crystallization. The rarity and partial resorption of high-Mg# *cpx* are consistent with dissolution during decompression and ascent (Neave and Mclennan 2020), which is a common process in arc magmas experiencing polybaric fractionation (Marxer et al. 2022). The presence of *opx* rims around *ol* may also indicate transfer of *ol* crystals to a more evolved (Si-rich) magmatic environment, since there is no other evidence of continuous evolution to a peritectic reaction boundary reaction (e.g., Yoder and Tilley 1962).

Geochemical evidence of open-system processes

In accordance with the mineral chemistry and textures discussed above, bulk rock geochemical evidence supports that the erupted compositions in Antimilos likely result from open-system magmatic processes. The linear trends between major element oxides in Harker diagrams (Fig. 4b–e), are not consistent with pure fractional crystallization and rather result from mixing between mafic and felsic components. Regarding the mafic endmember, even the least evolved LPBA (Mg# 54.5) is unlikely to be a potential candidate, since the low compatible trace element concentrations and the presence of relatively evolved olivine ($Fo_{70\text{-}85}$) antecrysts indicate that even the most primitive compositions collected on Antimilos have undergone fractionation. Additional evidence from the LPBA, such as high Al_2O_3 (18.4–18.7 wt%) and Sr (458–505 µg/g) contents and modest Eu anomalies (Eu/Eu* ~ 0.84–0.85), further indicate that fractionation occurred mainly outside of the *pl* stability field, or insufficient separation of crystallizing *pl*. While decrease in compatible trace element contents and the MREE-HREE abundances from the LPBA to HPAD (Fig. S2a–c) groups are consistent with fractionation of the observed mafic phases (*cpx, opx, mt*), a simple fractional crystallization scenario cannot account for the incompatible behavior of Sr and LREE enrichments in the HPAD. Therefore, an assimilation plus fractional crystallization (AFC) scenario looks like the most plausible option for the evolution of Antimilos suite, since we observe geochemical evidence of both fractional crystallization and mixing/assimilation.

MELTS: modeling the assimilation plus fractional crystallization scenario

We used the alphaMELTS software (Smith and Asimow 2005) to implement the rhyoliteMELTS 1.2 model of Ghiorso and Gualda (2015) to examine the relationship between observed major element trends and crystallization conditions. A pure FC scenario, of the most primitive LPBA sample (AM-10) tested under variable conditions, failed to reproduce the major element trends, as expected, given the widespread evidence from open system processes in Antimilos rocks. Therefore, we employed various (P, H_2O, fO_2) assimilation-fractional crystallization (AFC) scenarios to investigate whether the observed compositions result from combined crystallization and interaction of mafic magmas with residing felsic components. No samples mafic enough to be primary are present at Antimilos, so we selected as a starting point an Al-rich primitive basalt from elsewhere in the Aegean (specifically, the average basalt from the Pausanias area; Woelki et al. 2022) and two potential assimilants: (1) a matrix mineral (such as olivine or plagioclase), or (2) an evolved (rhyolitic) magma (sample M-115; Fytikas et al. 1986). The first scenario, which involved assimilation of minerals (olivine or plagioclase) into the fractionating melt, resulted in an excess of olivine/plagioclase at an early stage of evolution, which deviates significantly from the observed mineral assemblage, making it an unlikely process. The second scenario, involving the fractional crystallization of a basaltic melt incorporating a Si-rich component, resulted in liquid lines of descent at a broad range of conditions that matched well with the observed data for most of

the oxides (MgO, CaO, and TiO_2) versus SiO_2 (Fig. 4b, d, e). However, the trend in Al_2O_3 versus SiO_2 is quite sensitive to fractionation conditions and requires a specific combination of fO_2, pressure and water content to match the observed trend. These findings indicate that a reasonable model, thermodynamically constrained and satisfying all mass balance requirements, derives the Antimilos magmatic suite by fractionation at the QFM buffer (slightly more oxidizing than mid-ocean ridge basalt, but reduced compared to continental arcs), 100–400 MPa pressure, and 2.5 wt% initial water content, coupled with assimilation of a rhyolitic magma (Fig. 4). We note that the chemical evolution does not constrain whether the felsic component is a rhyolitic melt, or a mush, as textural evidence and mineral chemistry presented above support both scenarios.

The role of amphibole: conspicuously absent or well hidden?

Despite its absence from the erupted cargo of many arc magmas, whole-rock geochemical indices (Davidson et al. 2007) and observations in arc root complexes (Dessimoz et al. 2012) highlight the importance of amphibole in the petrogenesis of arc magmas. In Nisyros, for example, at the SE end of the Aegean arc, Klaver et al. (2017) interpreted amphibole-bearing mafic-to-ultramafic cumulates because of hydrous melt crystallization in the lower crust. These cumulates are thought to highlight the importance of *amp* in the production of silicic liquids via peritectic reactions (Blatter et al. 2017; Klaver et al. 2018) in deep crustal hot zones (Annen et al. 2006). A similar process has been proposed for magmas erupted on Milos (Zhou 2021; unpublished PhD Thesis) where modal amphibole is a significant component of the erupted products. In experimental studies of basaltic magmas from the Aegean arc, ferro-tschermakite to pargasitic hornblende compositions are stable up to 975 °C and $H_2O_{melt} \geq 3.5$ wt% at $fO_2 \sim$ NNO (Andujar et al. 2015). This upper temperature limit is substantially lower than the upper limit for *amp* (~1050 °C) in more primitive arc magmas (e.g., Pichavant et al. 2002; Grove et al. 2003; Krawczynski et al. 2012). The well-established tendency of amphibole to decompose due to fluctuations in T or P (Rutherford and Hill 1993; Rutherford and Devine 2003; Browne and Gardner 2006; DeAngelis et al. 2015) may account for the paucity of amphibole in the crystal record of Antimilos magmas. In this case, it would be expected that a textural record of *amp* decomposition might be preserved, as reported from nearby Milos (Xydous et al. 2021). However, no such record is apparent in Antimilos, and when coupled with our P–T-X_{H2O} estimates and trace element trends in low-Mg# *cpx* phenocrysts, one can argue that magmatic evolution and phenocryst growth in Antimilos took place in an environment where relatively low H_2O concentration and high storage temperatures inhibited amphibole crystallization. However, as argued above, we find it likely that amphibole was involved in the magmatic processes in a magmatic environment, deeper (mid-lower crust) within the plumbing system, whether it is still unclear if mafic recharges originate from this deeper environment or uptake antecrysts during ascent. Decreasing Dy/Yb and Dy/Dy* with SiO_2 observed in Antimilos bulk rocks is considered as an indicator of cryptic amphibole fractionation (Davidson et al. 2007; 2013). Furthermore, the amphibole signatures in resorbed high-Mg# *cpx* antecrystic cores (Fig. 9) are consistent with this scenario, which is expected given the sufficiently demonstrated presence of *amp* in arc root complexes and cumulate suites (Arculus and Wills 1980; Dessimoz et al. 2012; Smith 2014; Cooper et al. 2016; Klaver et al. 2017; Wang et al. 2019).

Synthesis and evolution of the Antimilos plumbing system

Decoupling between geochemistry and petrography is a common observation in continental volcanic arcs (Annen et al. 2006), often explained with reference to trans-crustal plumbing systems in which erupted magmatic products have petrographic features inherited during upper crustal magma chamber processes (Eichelberger 1978; Bachmann and Huber 2016). Our data and modeling suggest that the evolution of Antimilos parental magmas can be traced at least down to mid-crustal depths within the Aegean crust (~25 km; Tirel et al. 2004), likely outside of the plagioclase stability field. Primary melts at these levels evolved by *ol* + *cpx* + *amp* crystallization. The *amp* fractionation is cryptic (Davidson et al. 2007), either due to resorption during adiabatic ascent of near-liquidus magmas (e.g., Smith 2014) or to rapid destabilization upon mixing with drier magma. Either model would also account for the partially resorbed *cpx* antecrysts (Neave and Maclennan 2020; Marxer et al. 2022).

Lacking detailed geochronological data, inferences about the temporal evolution of the magmatic suite based on field and textural observations are tentative and subject to change. Given that HPAD units are stratigraphically below LPA and LPBA (Marinos 1961) and exhibit increased abundance of felsic antecrysts, we suggest that the cold and compositionally evolved ME-A was the dominant pre-HPAD feature in the Antimilos plumbing system. The origin of this residing evolved component is currently unresolved, but derivation from peritectic reactions involving amphibole, in a similar manner to Nisyros high porphyricity rhyodacite units (Klaver et al. 2017; 2018), is a potential scenario given the ubiquitous amphibole signatures in whole rock and *cpx* antecrysts trace element contents. This environment was rejuvenated and partially assimilated by ascending recharges (see

Fig. 10), which in turn evolve via the fractionation of the low-pressure assemblage $pl+cpx+opx\pm mt$. Mafic enclaves and abundant felsic antecrysts with variable degrees of reheating-induced disequilibrium textures suggest an open system process during the eruption of the highly porphyritic and mingled HPAD domes. Following HPAD volcanism, we propose that the more mafic parts of the plumbing system (ME-B) were tapped, indicated by the lower abundance felsic antecrysts and higher abundance of cognate mafic clots in LPA and LPBA. Unlike HPAD, the felsic clots in LPBA and LPA are completely reacted out and low-T phases, are rarely preserved as relics. Inter- and intra-unit variations in the degree of preservation of felsic antecrysts may result from different timing of entrapment in the mafic melt, as well as low dissolution rates (Laumonier et al. 2014). Flushing of resident mush piles by ascending melts and predominance of cognate material in the erupted products is considered a common process even in volcanoes erupting homogenous compositions through time (Streck et al. 2005; Wanke et al. 2019; Stock et al. 2020). Formation of phenocrysts and cognate clots likely took place in a post-recharge equilibration phase, driven by thermal mixing or ascent, as indicated by X_{An}-FeO systematics in phenocrystic and clot *pl*. On the other hand, sieve-textured *pl* displays steep X_{An}-FeO trends in line with compositional mixing. Microscale evidence of mingling with evolved felsic melts (rhyolite schlieren; Fig. S1i; Fig S7), even in the most primitive LPBA, further suggests that all the studied units evolved through variable degrees of interaction between mafic and felsic components.

Comparison with Milos

A range of compositions from basaltic andesites to rhyolites has been erupted over 3 Ma of volcanic history at Milos Island (Fytikas et al. 1986). This diversity of magmatic products in the MVF indicates open-system processes both in deep crustal hot zones (Zhou 2021, unpublished PhD Thesis) and in upper crustal mushes (Xydous et al. 2021). The dominance of rhyolites since 1 Ma (Fytikas et al. 1986; Zhou et al. 2021) points to a mature arc system in which alternating periods of volcanism and plutonism are dictated by changes in the magma supply rates in response to changes in the regional stress field (Zhou et al. 2022). Such processes are consistent with a major role for recharge filtering (Kent et al. 2010), causing mafic eruptions to eventually cease (Xydous et al. 2021).

On the other hand, Antimilos magmas reveal higher temperature phenocryst cargoes and geochemical evidence

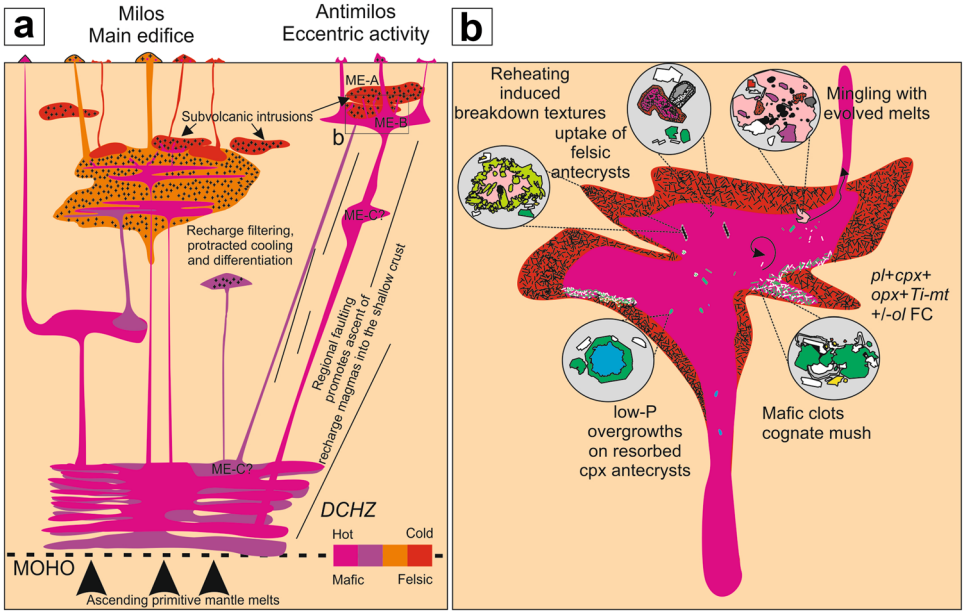


Fig. 10 Schematic illustration of the plumbing system of Antimilos, compared to the central edifice of the Milos volcanic field (**a**). Recharge magmas beneath Milos are effectively filtered in the upper crust, where stalling promotes magma hybridization, cooling and further differentiation to rhyolitic compositions. Mafic compositions are rare in the erupted record, albeit common as mafic enclaves mingled in lava domes. In Antimilos, the connection of subaerial and submarine volcanic centers with regional extensional structures suggests structurally controlled pathways. Hot, mafic magmas ascended rapidly through the crust, and erupted after limited cooling and variable degrees of interaction with felsic components. **b** Relation of the disequilibrium textures in the crystal cargo. Mafic magmas ascended in the upper crust and evolved through combined fractionation of $pl+cpx+opx+mt\pm ol$ and mixing with crystals and melts from the residing mush. Antecrysts are common, exhibiting disequilibrium features ranging from phenocrystic overgrowths, sieve textures and even complete reheating induced breakdown

consistent with limited fractionation during upper crustal storage. In Milos, upper crustal rhyolites can usually find their way to the surface, whereas the evolved compositions from ME-A in Antimilos are apparently unlikely to erupt without being remobilized by mafic recharge. Preferential eruption of andesites containing antecrysts from both mafic and felsic endmembers indicate a role for recharge filtering, but not as effective as it is at Milos. The proximity of Antimilos to the major WNW-ESE submarine extensional graben (Nomikou et al. 2013) may have promoted magma ascent through structurally controlled pathways, without prolonged shallow-level interactions. Mafic recharge bypasses, promoted by faulting, have been proposed to explain satellite eruptions in several arcs worldwide (e.g., Cassidy et al. 2015) and the basaltic eruptions of Pausanias adjacent to the Methana Volcanic Field (Woelki et al. 2022) in the Aegean arc. Another causative factor may involve different processes in the upper crustal storage regions of Milos and Antimilos (Fig. 10a). Cold, low-Si amphiboles are dominant in the erupted phenocryst, antecryst and mush fragment record of Milos (Xydous et al. 2021) between 2.36 and 1.04 Ma (ages from Zhou et al. 2021). Crystallization of *amp* increases magma crystallinity, promoting rheological lockdown, SiO_2 enrichment in the residual melts, and evolved melt extraction (Sisson and Grove 1993; Barclay and Carmichael 2004; Pichavant and MacDonald 2007; Nandenkar et al. 2014; Marxer et al. 2022). As already suggested for Methana (Elburg et al. 2018), we note here that a similar process is plausible for Milos, where crystallization of *amp* + *pl* in the upper crust may have formed extensive, rheologicaly locked mush zones that account for the predominance of erupted silicic compositions. At Antimilos, crystallization of amphibole during upper crustal storage is unlikely due to T-a_{H2O} conditions outside the *amp* stability field. Hence, the influence of amphibole on magma rheology is absent at Antimilos, allowing ascent and eruption of mafic recharges (Fig. 10b) instead of cooling and differentiation to more evolved liquids.

Conclusions

In this study, we combined petrographic observations, bulk rock compositions, and mineral major- and trace element data to investigate a suite of lavas collected from the understudied Antimilos volcano, located on the flank of the Milos Volcanic Field at the western end of the South Aegean Volcanic Arc. Our major findings are: (1) The lavas can be categorized into three suites based on their petrographic and compositional differences (less porphyritic basaltic andesite, LPBA; less porphyritic andesite, LPA; and highly porphyritic andesite–dacite, HPAD) but they are genetically related and represent members of a single magmatic plumbing system. These lavas were derived from magma that predominantly evolved in a lower-crustal magmatic system with minimal storage or evolution in the upper crust. Differences in chemical and petrographic features between the groups mostly reflect variations in the proportion of antecryst uptake and felsic melt assimilation. (2) Identification of high-Mg# clinopyroxene antecrysts showing evidence of crystallization alongside amphibole indicates the existence of a deep, hydrous reservoir beneath Antimilos, despite the nearly complete erasure of amphibole during subsequent magmatic evolution. (3) The presence of reacted felsic antecrysts in LPA and HPAD implies prior crystallization in a cold rhyolitic mush which did not effectively filter the eruption of relatively mafic lavas on Antimilos, but rather assimilated into the evolving mafic magma and affected its liquid line of descent. (4) The main edifice of the Milos Volcanic Field shows a plumbing system distinct from the satellite system active at Antimilos, with a greater role for recharge filtering, upper crustal cooling, hybridization, and ongoing amphibole fractionation. The sampling of a hot crystallization path at Antimilos may be linked to its peripheral location within the MVF and nearby extensional structures.

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Data availability All the available data and figures supporting our findings are given in the text and supplementary files.

Declarations

Conflict of interest The authors have no financial or proprietary interests in any material discussed in this article.

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